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HISTORY · Bhakti and Sufi movements

UNDERSTAND IT FIRST

What is it? Bhakti means **loving devotion** to a personal God; **Sufism** is the **mystical** side of Islam, the search for **union with God through love**. Both grew in medieval India between about the **6th and 17th centuries**, when rigid ritual, caste and priestly control troubled ordinary people. Both said that **God can be reached through love, not through ceremony, birth or learning**, both used **local languages** and **music**, and both shaped Indian society, language and **religious harmony**. Learn it as **Bhakti (south → north → west) → Sufism (orders, ideas) → Sikhism → Impact**.

The Bhakti movement

1. Bhakti in the south (6th–13th century)

Group	Facts
Alvars (Vishnu)	12 saints; hymns in Tamil collected as the **Nalayira Divya Prabandham**. **Nammalvar**, **Tirumangai**, **Periyalvar**, **Andal** (the woman Alvar, sang of her love for Vishnu: *Tiruppavai*).
Nayanars (Shiva)	63 saints; hymns collected as the **Tevaram** (**Appar, Sambandar, Sundarar**) and **Tirumurai**; life stories in **Periyapuranam** (Sekkizhar). **Karaikkal Ammaiyar**, a woman saint.
Philosophers (acharyas)	**Shankara** (8th c., **Advaita**, non-dualism); **Ramanuja** (11th–12th c., **Vishishtadvaita**, qualified non-dualism, *Sri Bhashya*, Sri Vaishnavism); **Madhva** (13th c., **Dvaita**, dualism); **Nimbarka** (**Dvaitadvaita**); **Vallabha** (**Shuddhadvaita**, Pushtimarg).
Basavanna (12th c., Karnataka)	Minister of **Bijjala** at **Kalyana**; founded the **Veerashaiva / Lingayat** movement; **Anubhava Mantapa** (hall of experience); wrote **Vachanas** in Kannada; opposed **caste, idol worship and rituals**; saints **Akka Mahadevi**, **Allama Prabhu**.

2. Bhakti in the north

Saint	Facts
Ramananda (14th–15th c., Banaras)	Devotee of **Rama**; taught in **Hindi**, not Sanskrit; accepted disciples of **all castes**: **Kabir** (weaver), **Ravidas** (cobbler), **Sena** (barber), **Dhanna** (Jat), **Pipa** (Rajput).
Kabir (c. 1440–1518)	**Nirguna** (formless) God; **Hindu–Muslim unity**; attacked **idol worship, caste, pilgrimage, empty ritual**; compositions: **dohas, sakhis, sabads, ramainis**, collected in the **Bijak**; followers: **Kabirpanthis**.
Ravidas (Raidas)	Cobbler-saint of Banaras; **nirguna** devotion; verses included in the **Guru Granth Sahib**; **Ravidassia** community.

Guru Nanak (1469–1539)	Born at **Talwandi** (Nankana Sahib); **one formless God (Ik Onkar)**, *Japji*; rejected **idol worship, caste and rituals**; started **langar** (community kitchen), **sangat** (congregation) and **pangat** (eating together); settled at **Kartarpur**; made **udasis** (journeys).
Chaitanya (1486–1534)	Born at **Nadia (Bengal)**; **Krishna** devotion (**Gaudiya Vaishnavism**); **sankirtan** (congregational singing); **Achintya Bheda-Abheda** philosophy; spread the movement to **Puri (Odisha)** and **Vrindavan**.
Tulsidas (1532–1623)	Banaras; **Rama** devotion (**saguna**); **Ramcharitmanas** in **Awadhi** (Akbar's time); also *Vinaya Patrika*, *Kavitavali*.
Surdas (c. 1483–1563)	Blind poet; **Krishna** devotion; **Sur Sagar**; disciple of **Vallabha**; **Braj** language; lived near Agra.
Mirabai (16th c.)	Rathor princess of **Merta (Rajasthan)**, wife of Bhoj Raj of **Mewar**; devotion to **Krishna**; songs in **Rajasthani/Braj**; defied family custom; considered **Ravidas** her guru.
Sankaradeva (1449–1568, Assam)	**Ekasarana dharma** (devotion to one God, Krishna); **Namghar** (prayer hall); **Kirtan Ghosa**; **Ankiya Nat** (one-act plays); **Sattras** (monasteries).

3. Bhakti in the west, south and Andhra

Region	Saints and facts
Maharashtra (**Varkari** tradition; **Vithoba of Pandharpur**)	**Jnaneshwar** (*Dnyaneshwari*, 1290), **Namdev** (tailor), **Eknath**, **Tukaram** (*abhangs*), **Janabai**; **Ramdas** (*Dasbodh*), guru of Shivaji.
Gujarat, Karnataka	**Narsinh Mehta** (*Vaishnav Jan To*); **Purandaradasa**, **Kanakadasa** (Haridasa tradition).
Andhra and Telangana	**Annamacharya** (Tirumala, Venkateswara sankirtanas); **Potana** (*Bhagavatam*); **Kancherla Gopanna (Ramadasu)** at **Bhadrachalam** (Rama devotion); **Kshetrayya** (Krishna padams, Muvva Gopala); **Tyagaraja** (Telugu kritis, Thanjavur); **Vemana** (verses against caste and ritual, 17th c.); **Pothuluri Veerabrahmam** (*Kalagnanam*, Kadapa region); Ramanuja's Sri Vaishnava link with **Tirupati**; Veerashaiva and Haridasa influence in the Rayalaseema–Karnataka border.

4. Teachings of Bhakti

- **One God**, reached through **love and surrender**, not rituals, sacrifices or priests. **Nirguna** (God without form: Kabir, Nanak) and **Saguna** (God with form: Rama, Krishna: Tulsidas, Surdas, Mirabai).
- **Against caste and untouchability**; saints came from **all castes**; **women** saints (Andal, Akka Mahadevi, Mirabai, Janabai); the **guru** is essential; singing (**kirtan**, **bhajan**) in the **local language**.

5. Sufism

Main Sufi orders

Term	Meaning
Silsila	An **order** or chain of teachers and disciples; **12–14 silsilas** reached India (Chishti, Suhrawardi, Qadiri, Naqshbandi, Firdausi, Shattari and others).
Pir–Murid, **Khanqah**	Master and disciple; the **khanqah** is the hospice where the Sufi lives, teaches and feeds the poor.
Sama, **Zikr**	**Sama** = devotional music and song (qawwali); **zikr** = repeating God's name.
Wahdat-ul-wujud	'Unity of being' (**Ibn Arabi**): all is God. **Shaikh Ahmad Sirhindi** (Naqshbandi) opposed it and taught **wahdat-ush-shuhud** (unity of witness), and is called **Mujaddid Alf-i-Thani** (renewer of the second millennium).
Fana, **Baqa**	**Fana** = losing the self in God; **baqa** = living on in God.
Malfuzat, **Urs**	**Malfuzat** = recorded sayings (e.g. *Fawaid-ul-Fuad* of **Nizamuddin Auliya** by **Amir Hasan Sijzi**); **urs** = death anniversary of a saint, a festival at the **dargah**.

Order	Saints and features
Chishti	Brought by **Khwaja Moinuddin Chishti** (Ajmer, died 1236); **Qutbuddin Bakhtiyar Kaki** (Delhi); **Baba Farid** (Pakpattan; verses in the Guru Granth Sahib); **Nizamuddin Auliya** (Delhi, died 1325, disciple **Amir Khusrau**); **Nasiruddin Chiragh-i-Delhi**; **Gesu Daraz** (Gulbarga). Lived **simply**, kept away from rulers and **refused state posts and grants**, loved **music (sama)**; **Salim Chishti** of Fatehpur Sikri blessed Akbar.
Suhrawardi	**Shaikh Bahauddin Zakariya** (Multan), **Hamiduddin Nagauri**; **accepted government posts and grants** and worked closely with the Sultans; strong in **Sind and Punjab**.
Qadiri	Founded by **Abdul Qadir Jilani**; came to India in the **15th century** (Uch, Sind); **Mian Mir** (Lahore); **Dara Shikoh** was a follower. Popular in **Punjab, Kashmir and the Deccan**.
Naqshbandi	Came with **Khwaja Baqi Billah** (under Akbar); **Shaikh Ahmad Sirhindi** stressed the **sharia** and opposed Akbar's liberal ideas; later **Shah Waliullah**.
Rishi order (Kashmir)	**Sheikh Nuruddin Nurani (Nund Rishi)** and the Shaiva mystic **Lal Ded (Lalla)**, who composed **Vakhs**, joined Sufi and Shaiva ideas.
Sufi shrines in Andhra	**Kadapa** (Ameen Peer / Pedda Dargah), **Nellore** (Barah Shaheed Dargah), **Penukonda** (Baba Fakruddin Dargah): major dargahs where Hindus and Muslims both attend urs.

6. Sikhism: the ten Gurus

Guru	Contribution
1 **Nanak**	Founder; one God; langar, sangat, pangat.
2 **Angad**	Developed the **Gurmukhi** script.

3 **Amar Das**	Strengthened **langar**; opposed **sati** and purdah; **manji** system of preaching; Akbar visited his langar.
4 **Ram Das**	Founded **Amritsar**; Akbar granted the land.
5 **Arjan**	Built **Harmandir Sahib**; compiled the **Adi Granth (1604)**; **executed by Jahangir (1606)** (the first Sikh martyr).
6 **Hargobind**	**Miri–Piri** (temporal and spiritual power); army; **Akal Takht**.
7 **Har Rai**, 8 **Har Krishan**	Consolidation; Har Krishan was a child Guru.
9 **Tegh Bahadur**	**Executed by Aurangzeb (1675)** in Delhi for defending Kashmiri Pandits' religious freedom.
10 **Gobind Singh**	Founded the **Khalsa (1699)** at Anandpur with the **five Ks** (kesh, kangha, kada, kirpan, kachera); declared the **Guru Granth Sahib** the eternal Guru.

7. Bhakti and Sufi compared; impact

Point	Bhakti	Sufism
Aim	Devotion to a personal God	Union with God through love
Teacher	Guru	Pir (murid–pir)
Music	Kirtan, bhajan, abhang	Qawwali, sama
Shrine	Temple, namghar, Nam-kirtan	Dargah, khanqah
Common	Love, equality, against empty ritual, local language, harmony	Same

Area	Impact
Politics	Eased tension; supported **Akbar's Sulh-i-kul**; but **Sikh–Mughal conflict** grew into a political force (Khalsa).
Society	Challenged **caste**; saints from **artisan and low castes** (Kabir weaver, Ravidas cobbler, Namdev tailor, Sena barber); status of **women** saints.
Economy	Saints often earned by **trade and crafts** (dignity of labour); **langar** fed the poor; dargahs received **endowments** and drew **pilgrims and trade**.
Religion, language, culture	Birth of **Sikhism**; growth of **Hindi, Bengali, Marathi, Telugu, Kannada, Assamese, Punjabi**; new music and devotional literature.

EXAM TRAP

Alvars = Vishnu, **Nayanars** = Shiva; **Ramanuja**: Vishishtadvaita; **Madhva**: Dvaita; **Shankara**: Advaita; **Basavanna**: Vachanas; **Ramananda** → Kabir, Ravidas; **Kabir**: Bijak; **Nanak**: langar; **Chaitanya**: Nadia, sankirtan; **Tulsidas**: Ramcharitmanas; **Sankaradeva**: Namghar; **Chishti**: Moinuddin (Ajmer), Nizamuddin;

Suhrawardi accepted state posts; **Naqshbandi**: Sirhindi; **Arjan**: Adi Granth 1604, executed 1606; **Gobind Singh**: Khalsa 1699.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the Bhakti movement:
1. Ramananda accepted disciples of different castes, including Kabir and Ravidas.
2. Tulsidas composed the Ramcharitmanas in Awadhi.
3. Chaitanya was born in Maharashtra and popularised the Varkari tradition.
4. Basavanna founded the Veerashaiva movement in Karnataka.
Which of the above are correct?

A. 1, 2 and 4 only

B. 1 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. Chaitanya was from Nadia in Bengal and preached Krishna bhakti; the Varkari tradition belongs to Maharashtra.

PRACTICE

Match List-I (Sufi order) with List-II (saint):
List-I: A. Chishti B. Suhrawardi C. Naqshbandi D. Qadiri
List-II: 1. Shaikh Ahmad Sirhindi 2. Nizamuddin Auliya 3. Mian Mir 4. Bahauddin Zakariya

A. A-2, B-4, C-3, D-1

B. A-4, B-2, C-1, D-3

C. A-2, B-4, C-1, D-3

D. A-2, B-1, C-4, D-3

Answer: C. Chishti: Nizamuddin; Suhrawardi: Zakariya; Naqshbandi: Sirhindi; Qadiri: Mian Mir.

PRACTICE

Which one of the following statements is NOT correct?

A. Guru Arjan compiled the Adi Granth in 1604 and was executed in 1606.

B. The Suhrawardi saints accepted government posts and grants, unlike the Chishtis.

C. Shaikh Ahmad Sirhindi opposed wahdat-ul-wujud and stressed the sharia.

D. The Chishti saints readily took official posts and sought close ties with the Sultans.

Answer: D. The Chishtis kept away from rulers and refused state posts; the Suhrawardis accepted them.

PRACTICE

Assertion (A): The Bhakti movement helped the growth of regional languages.
Reason (R): Bhakti saints preached in the language of the common people rather than in Sanskrit.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Kabir, Tulsidas, Tukaram, Annamacharya and others wrote in the people's own languages.

PRACTICE

A saint of Assam founded the Ekasarana dharma, set up prayer halls called Namghars and wrote the Kirtan Ghosa. Who was he?

A. Chaitanya

B. Sankaradeva

C. Vallabhacharya

D. Eknath

Answer: B. Sankaradeva (1449–1568) of Assam.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the Bhakti saints region-wise with one fact each. 2. Write the four Sufi orders, their saints and how they differ. 3. Write the ten Gurus and the Bhakti–Sufi comparison.

POLITY · Parliament: composition, powers, Bills and the Budget

UNDERSTAND IT FIRST

What is it? Parliament is the **law-making and money-controlling body** of the Union. This day goes deeper than Day 9: **who can sit in it**, **how it works** (sessions, motions, committees), **how a Bill becomes law**, and **how the Budget is passed**. The big idea: **the government can spend or tax only if Parliament permits it** (Art 265: *no tax except by authority of law*; Art 266: no money from the Consolidated Fund except by law).

1. Composition, membership and the Speaker

Point	Details
Strength	**Lok Sabha: maximum 550** (530 from States, 20 from UTs) after the **104th Amendment (2019)** ended Anglo-Indian nomination (earlier 552); **543** elected. **Rajya Sabha: maximum 250** (238 + 12 nominated). Seats are fixed by the **1971 Census** until the first Census after 2026 (42nd, 84th Amendments); seats are **reserved for SCs and STs** (Art 330).
Qualifications (Art 84)	Citizen of India; **25 years** (Lok Sabha), **30 years** (Rajya Sabha); other qualifications by the **Representation of the People Act, 1951**; must **take an oath**.
Disqualification (Art 102)	**Office of profit** under the Government; **unsound mind** declared by a court; **undischarged insolvent**; **not a citizen** or acquired foreign citizenship; **disqualified by law**; and for **defection** under the **Tenth Schedule**.
Anti-defection (Tenth Schedule)	Added by the **52nd Amendment (1985)**; a member is disqualified if he **voluntarily gives up** party membership or **votes against the party whip**; the **Presiding Officer** decides; **Kihoto Hollohan (1992)** held that the decision can be reviewed by courts; the **91st Amendment (2003)** removed the **'one-third split'** exception and kept only the **two-thirds merger**.
Speaker (Art 93–96)	Elected by the Lok Sabha from its members; **presides**, keeps order, **decides if a Bill is a Money Bill (final, Art 110(3))**, **casting vote only in a tie**; resigns to the **Deputy Speaker**; removed by a resolution of **all the then members** (effective majority) with **14 days' notice**; a **pro tem Speaker** administers oaths at the start.

2. How Parliament works

Feature	Details
Sessions (Art 85)	Summoned by the **President**; **gap between two sessions not more than 6 months**; three sessions: **Budget** (Feb–May), **Monsoon** (Jul–Aug), **Winter** (Nov–Dec); **quorum = 1/10** of members (Art 100).
Question Hour, Zero Hour	**Question Hour** (first hour; starred, unstarred, short notice questions) is a **control over the executive**; **Zero Hour** (after Question Hour) is an **Indian innovation** (not in the rules) for urgent matters.

Motions	**Adjournment** (urgent public matter; **only Lok Sabha**; 50 members' support); **Calling attention**; **No-confidence** (only Lok Sabha; 50 members' support); **Censure**; **Privilege**; **Cut motions** (policy cut Re 1, economy cut, token cut Rs 100) on demands for grants.
Committees	**Public Accounts Committee** (22: 15 LS + 7 RS; **chaired by an Opposition member** by convention; checks **CAG reports**); **Estimates Committee** (30, only LS); **Committee on Public Undertakings** (22); **Departmentally Related Standing Committees** (24, from **1993**; 31 members each); **Select** and **Joint** committees.
Privileges	**Freedom of speech** in Parliament (Art 105), **immunity** from court proceedings for what is said or voted; **arrest** protection in civil cases during session; **contempt** powers.

3. Bills: kinds and passage

How an ordinary Bill becomes law

Kind	Rules
Ordinary Bill (Art 107–108)	Introduced in **either House**; needs a **simple majority**; **Rajya Sabha** can amend or reject; deadlock: **joint sitting (Art 108)**; the **President** may assent, withhold, or **return once**.
Money Bill (Art 110)	Deals **only** with **taxes, borrowing, the Consolidated or Contingency Fund, appropriation** and similar matters; **only in Lok Sabha**, on the **President's recommendation**; **Speaker certifies** it (final); **Rajya Sabha** may only **recommend changes within 14 days**; Lok Sabha may **accept or reject** them; if RS does not return it in 14 days it is **deemed passed**; **no joint sitting**; the President **cannot return** it (can only assent or withhold).
Financial Bill (Art 117)	**Category I**: has money provisions **plus** other matters; only in **LS** with President's recommendation; **joint sitting** possible. **Category II**: only involves **expenditure from the Consolidated Fund**; **either House**; cannot be passed **unless the President recommends** consideration.
Constitution Amendment Bill (Art 368)	Either House; **special majority**: **majority of the total membership** of each House **and 2/3 of members present and voting**; for **federal features** also **ratification by at least half of the States**; **no joint sitting**; the President **must assent** (24th Amendment). **Kesavananda Bharati (1973)**: Parliament cannot change the **basic structure**.

Stages of an ordinary Bill: (1) **First reading**: introduction. (2) **Second reading**: general discussion, then **committee stage** (clause by clause), then **report stage**. (3) **Third reading**: final vote. Then the **other House** repeats the stages, and the **President** gives assent. A Bill passed by Parliament lapses only if it is **pending in the Lok Sabha** when it is dissolved; a Bill pending in the Rajya Sabha but **not passed by the Lok Sabha** does **not** lapse.

4. The Budget

The Union Budget

Term	Meaning
Annual Financial Statement (Art 112)	The **Budget**: estimated receipts and expenditure for the year (1 April to 31 March); shows **expenditure charged** on the Consolidated Fund separately from other expenditure. Presented in the **Lok Sabha** on **1 February** (since 2017; the **Railway Budget** was merged in 2017).
Consolidated Fund (Art 266(1))	All **revenues, loans and repayments** of the Government; money can be drawn **only by an Appropriation law**.
Contingency Fund (Art 267)	A **'float'** (corpus fixed by law) kept at the **President's disposal** for **unforeseen expenditure**, later reimbursed by Parliament; the **Public Account** (Art 266(2)) holds provident funds, small savings, deposits.
Charged expenditure	Items that are **discussed but not voted**: salary of the **President**, **judges**, **CAG**, **Speaker**, interest and debt charges. (Votable items: **demands for grants**.)
Demands for Grants (Art 113)	Voted **only in the Lok Sabha**; the Rajya Sabha only discusses. **Guillotine**: remaining demands passed together without discussion when the time ends.
Appropriation Bill (Art 114)	Authorises withdrawal from the Consolidated Fund; no money can be withdrawn without it.
Finance Bill	Gives effect to the **tax proposals**; a **Money Bill** (Art 110(1)(a)); must be passed **within 75 days**.
Vote on account (Art 116)	**Advance grant** to meet expenses while the full Budget is pending (usually **2–4 months**, often in an **election year**); other grants: **supplementary, additional, excess, vote of credit, exceptional**.
Deficits	**Revenue deficit** = revenue expenditure − revenue receipts; **Fiscal deficit** = total expenditure − total receipts **excluding borrowings** (the amount the Government must borrow); **Primary deficit** = fiscal deficit − interest payments. **FRBM Act, 2003** sets targets.

EXAM TRAP

LS max 550 (104th Amendment); **age 25 / 30**; **Art 102** disqualification; **Tenth Schedule** (52nd, 91st); **gap ≤ 6 months**; **quorum 1/10**; **PAC chaired by Opposition**; **Money Bill: LS only, RS 14 days, no joint sitting, Speaker certifies**; **Art 368**: majority of total + 2/3 present and voting, half of States for federal features; **Art 112 Budget, 114 Appropriation, 116 Vote on account, 266 Funds, 267 Contingency Fund**; **Finance Bill within 75 days**; **Fiscal deficit = expenditure − receipts (without borrowings)**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about Bills in Parliament:

1. A Money Bill can be introduced only in the Lok Sabha.

2. The Speaker's certificate that a Bill is a Money Bill is final.

3. The Rajya Sabha can reject a Money Bill passed by the Lok Sabha.

4. A joint sitting can be called to resolve a deadlock on a Money Bill.

Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 3 only

C. 2, 3 and 4 only

D. 1, 3 and 4 only

Answer: A. Rajya Sabha can only recommend changes; there is no joint sitting for Money Bills.

PRACTICE

Match List-I (Article) with List-II (subject):

List-I: A. Article 112 B. Article 114 C. Article 116 D. Article 267

List-II: 1. Vote on account 2. Contingency Fund of India 3. Annual Financial Statement 4. Appropriation Bills

A. A-3, B-4, C-1, D-2

B. A-4, B-3, C-1, D-2

C. A-3, B-4, C-2, D-1

D. A-1, B-4, C-3, D-2

Answer: A. 112: Budget; 114: Appropriation; 116: Vote on account; 267: Contingency Fund.

PRACTICE

Which one of the following statements is NOT correct?

A. Charged expenditure on the Consolidated Fund is discussed but not voted.

B. The Speaker has a casting vote only when there is a tie.

C. The Public Accounts Committee is chaired by a member of the ruling party.

D. An amendment of the Constitution needs a majority of the total membership and two-thirds of members present and voting.

Answer: C. By convention the PAC is chaired by a member of the Opposition.

PRACTICE

Assertion (A): The Rajya Sabha cannot amend or reject a Money Bill.
Reason (R): A Money Bill can be introduced only in the Lok Sabha, and the Rajya Sabha can only make recommendations within 14 days.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Because the Rajya Sabha only recommends, its power over a Money Bill is limited, as R explains.

PRACTICE

The Lok Sabha passes a Money Bill and sends it to the Rajya Sabha, which keeps it for more than 14 days without returning it. What is the result?

A. The Bill lapses

B. The Bill is deemed passed by both Houses

C. A joint sitting is summoned

D. The Bill is sent to a Select Committee

Answer: B. Art 109: if the Rajya Sabha does not return a Money Bill within 14 days, it is deemed passed.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write qualifications, disqualifications and the Tenth Schedule. 2. Write the table of four kinds of Bills. 3. Draw the Budget chart and define Art 112, 114, 116, 266, 267 and the three deficits.

MATHS · Ratio and proportion

UNDERSTAND IT FIRST

What is it? A **ratio** compares two quantities of the **same kind** by division: if a class has 20 boys and 30 girls, boys : girls = 20 : 30 = **2 : 3**. It has **no unit**, and it tells **how many times** (or what part) one is of the other. A **proportion** says that **two ratios are equal**: 2 : 3 = 4 : 6. Almost all of **partnership, mixtures,**

ages, speed–time–work and percentages are ratio problems in disguise, so this topic is a base for the next several days.

Ratio and proportion: the toolkit

1. Ratio

- In **a : b**, **a** is the **antecedent** and **b** the **consequent**. Both quantities must be in the **same unit** (convert ₹ and paise, or metres and cm, first). Multiply or divide both terms by the same number to get an **equal ratio**; reduce to the **simplest form**.
- **Compare:** a : b > c : d if **ad > bc**. **Kinds: duplicate** $a^2 : b^2$; **triplicate** $a^3 : b^3$; **sub-duplicate** $\sqrt{a} : \sqrt{b}$; **inverse** b : a; **compound** ac : bd (multiply the antecedents and the consequents).
- **Dividing a quantity N in the ratio a : b** gives **aN/(a + b)** and **bN/(a + b)**. **Joining ratios:** if A : B = 2 : 3 and B : C = 4 : 5, make **B equal** (the LCM of 3 and 4 is 12): A : B = 8 : 12 and B : C = 12 : 15, so **A : B : C = 8 : 12 : 15**.
- **Adding the same number** to both terms moves the ratio **towards 1 : 1**: adding 1 to 3 : 5 gives 4 : 6 = 2 : 3 (0.667), nearer to 1 than 3 : 5 (0.6).

2. Proportion

- **a : b = c : d** (a, b, c, d in proportion): **a and d** are the **extremes**, **b and c** the **means**; **product of extremes = product of means**: **ad = bc**. This is the **cross-multiplication** rule used in every rule-of-three problem.
- **Continued proportion** a : b = b : c: $\mathbf{b^2 = ac}$, so $\mathbf{b = \sqrt{(ac)}}$ is the **mean proportional**. **Third proportional** to a and b: a : b = b : x gives $\mathbf{x = b^2/a}$. **Fourth proportional** to a, b, c: a : b = c : x gives **x = bc/a**.

Property (if a : b = c : d)	Result
Invertendo	b : a = d : c
Alternendo	a : c = b : d
Componendo	(a + b) : b = (c + d) : d
Dividendo	(a − b) : b = (c − d) : d
Componendo and dividendo	(a + b) : (a − b) = (c + d) : (c − d)

EXAMPLE

Example: if **(x + y) : (x − y) = 7 : 3**, apply componendo and dividendo: **2x : 2y = (7 + 3) : (7 − 3) = 10 : 4**, so **x : y = 5 : 2**. **Variation: direct** (y/x constant: more goods, more cost) and **inverse** (xy constant: more men, fewer days). **Partnership:** profit is shared in the ratio of **(capital × time)**. If A puts ₹30,000 for 8 months and B ₹40,000 for 6 months, the ratio is 240 : 240 = **1 : 1**.

EXAM TRAP

ad = bc; mean proportional = √(ac); third proportional = b^2/a; fourth proportional = bc/a; share = (part / total parts) × N; joining ratios: equalise the common term; componendo-dividendo: (a+b)/(a−b) = (c+d)/(c−d); partnership = capital × time.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

A : B = 2 : 3 and B : C = 4 : 5. If a sum of ₹3,500 is divided among A, B and C in the ratio A : B : C, what is C's share?

A. ₹1,200

B. ₹1,500

C. ₹1,800

D. ₹1,400

Answer: B. A : B : C = 8 : 12 : 15 (total 35); C = 15/35 × 3,500 = ₹1,500.

PRACTICE

If x is the mean proportional between 9 and 16, and y is the third proportional to 6 and 9, what is the value of x + y?

A. 24

B. 25.5

C. 26

D. 27

Answer: B. x = √(9 × 16) = 12; y = $9^2/6$ = 13.5; x + y = 25.5.

PRACTICE

The present ages of a father and his son are in the ratio 7 : 2. After 10 years the ratio will be 9 : 4. What is the father's present age?

A. 28 years

B. 35 years

C. 42 years

D. 49 years

Answer: B. (7k + 10) : (2k + 10) = 9 : 4 gives 28k + 40 = 18k + 90, so k = 5; father = 35.

PRACTICE

A invests ₹30,000 for 8 months and B invests ₹40,000 for 6 months in a business. If the total profit is ₹1,200, what is B's share?

A. ₹480

B. ₹600

C. ₹700

D. ₹720

Answer: B. 30,000 × 8 = 2,40,000 and 40,000 × 6 = 2,40,000: ratio 1 : 1, so B = ₹600.

PRACTICE

If (x + y) : (x − y) = 7 : 3, then x : y is

A. 5 : 2

B. 7 : 3

C. 4 : 3

D. 2 : 5

Answer: A. Componendo-dividendo: 2x : 2y = 10 : 4, so x : y = 5 : 2.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the ratio kinds and how to join A : B and B : C. 2. Write proportion, mean, third and fourth proportional and the five properties. 3. Solve one example each of sharing, ages and partnership.

NOTEBOOK PAGE · Day 11: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

HISTORY: Bhakti and Sufi movements

Area	Write
Bhakti: south	Alvars (Vishnu, Divya Prabandham, Andal), Nayanars (Shiva, Tevaram), Shankara, Ramanuja, Madhva, Basavanna (Vachanas)
Bhakti: north	Ramananda → Kabir, Ravidas; Nanak; Chaitanya; Tulsidas, Surdas, Mirabai; Sankaradeva; Maharashtra: Jnaneshwar, Namdev, Eknath, Tukaram
Teachings	One God, devotion over ritual, against caste, vernacular languages, guru, nirguna and saguna
Sufi orders	Chishti (Moinuddin, Kaki, Farid, Nizamuddin), Suhrawardi (Zakariya), Qadiri (Mian Mir), Naqshbandi (Sirhindi)
Sufi ideas	Pir–murid, khanqah, sama, zikr, wahdat-ul-wujud, fana, malfuzat, urs
Sikhism	10 Gurus; Nanak (langar, sangat, pangat), Arjan (Adi Granth 1604), Gobind Singh (Khalsa 1699)

POLITY: Parliament

- Composition and qualifications (25 / 30 years), Art 102 disqualifications, Tenth Schedule; sessions (gap ≤ 6 months), quorum 1/10; Speaker; committees (PAC, Estimates, COPU); motions.
- Bills: ordinary, money, financial, amendment (Art 368); Budget: Art 112, 114, 116, 266, 267; Appropriation Bill, Finance Bill; deficits.

MATHS: Ratio and proportion

Write: ratio, proportion (ad = bc), mean, third and fourth proportional, invertendo, alternendo, componendo, dividendo; dividing a quantity in a ratio; joining two ratios (A:B and B:C); partnership (capital × time); one worked example of each.